WeBERT-A (Weighted BERT with Augmentation)

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Github

GitHub Link: https://github.com/AAreeb13/NLP-Coursework

1 Stage 1: Task Definition and Literature Review

1.1 Q1: Primary Contributions

The PCL paper is about analysing the use of Patronising and Condescending Language towards vulnerable people in media. The paper identifies a gap in NLP research for detecting offensive language.

The key contribution of the paper is the 'Don't Patronize Me!' dataset, which contains 10,637 paragraphs on vulnerable social groups. The researchers used 10 keywords and 20 English speaking countries to build their dataset (extracted from News on Web Corpus) of rich-context data based on whether or not the keywords are seen in the paragraphs and set to annotate the whole dataset using three expert annotators.

The dataset is categorised with their own PCL categories taxonomy into seven different categories and three higher level categories:

1. **The saviour.** The author uses language that hints at having a superior position/privileged situation.
2. **The expert.** The author uses language that presumes better understanding of the targeted community
3. **The poet.** The author uses descriptive language to discredit the vulnerable communities situation by idealising or romanticise the situation.

Through multi-level annotation, the dataset features both paragraph-level identification and span-level categorisation of PCL. The annotation methodology is spoken about in detail.The first step is labelling how visible PCL is within the paragraph in question and produces a score between 0 and 5. The second step includes identifying the specific PCL's found based on the 7 categories previously mentioned.

1.2 Q2: Technical Strengths

The first strength of the paper is the internal scrutiny, visibility and care taken in the **annotation process**. Providing a detailed description of the process enables us as readers to transparently scrutinise the process. In this case, we can see that expertise are leveraged and their agreeability was measured with Kappa IAA. We observe that IAA achieved was 41% or 61% meaning that the task can be made to not be overly ambiguous, if framed correctly. We can also highlight the diversity of the data.

We have a vast amount of unprocessed data to explore. Approximately 10,000 datapoints, approximately evenly distributed across 20 countries and about an even spread of data for each of the keywords. As mentioned before, the clarity of the scope was slightly explored when mentioning 61% IAA score indicating it was possible to identify data classes/categories to a degree of ease. We can also note that an 'Other' category

for PCL data was not needed since the 7 categories provided enough breadth to fit each datapoint in at least one of them. Finally

The author also provides sufficient explanation to why the issue is significant. The paper is building upon current classification problems of ill-natured, however, tackles a niche (not a completely new idea), yet a more complex problem that is less explored in NLP.

1.3 Q3: Key Weaknesses

1.3.1 Limitations of the Dataset

The dataset is a key contribution; hence, it is important to mention its drawbacks:

- The researchers presented two values for IAA, 41% and 61%. 61% was only achievable if we removed all datapoints that were ambiguously classified or disagreements. This does highlight a significant degree of subjectivity by even expert annotators.
- We observe a good spread across the 10 keywords for their dataset. However, for their category taxonomy, we notice some inbalances in the data. For example, the category **merr** had only 64 while **com** had 888. This is a hugely uneven dataset. An even more concerning fact is the dataset used to test and train models included only 995 positive examples out of approximately 9,000. This provides significant bias to the trained model.

1.3.2 Limitations in Evaluation

The paper indicates that PCL is present at a text span level. Task 2 of the evaluation demonstrates that we only evaluated at a paragraph-level.

While the authors do categorise the data well, there is a lack of systematic error analysis regarding the models' false positives. For instance, the paper does not clarify whether the models are simply triggering on "vulnerable group" keywords such as refugee, rather than actually understanding the patronizing tone, which is a common pitfall in bias-detection tasks.

2 Stage 2: Exploratory Data Analysis

2.1 Basic Statistical Profiling

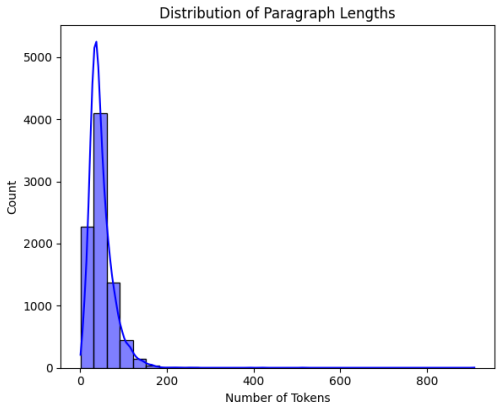
The following table summarizes the token distribution and vocabulary metrics for the dataset.

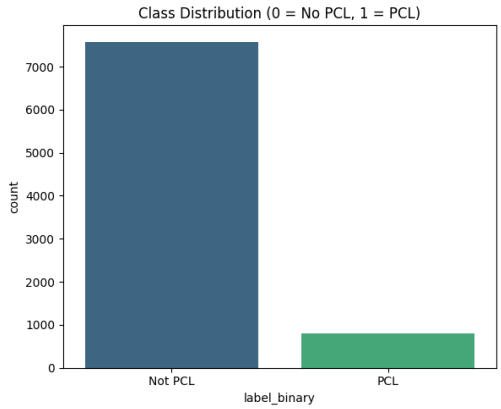
Metric	**Value**
Average Token Count	48.68
Minimum Token Count	1
Maximum Token Count	909
Vocabulary Size	29,540

Table 1: Token and Vocabulary Statistics

Class Distribution: The dataset consists of five distinct classes, distributed as follows:

- **Class 0:** 6,825 (81.49%)
- **Class 1:** 756 (9.03%)
- **Class 2:** 126 (1.50%)
- **Class 3:** 369 (4.41%)
- **Class 4:** 299 (3.57%)





2.1.1 Analysis

The dataset shows a significant right-skewed distribution in paragraph lengths. We see an average of 49 tokens but span from 1 to 900 in token sizes. We can also see that there is a severe imbalance in classes.

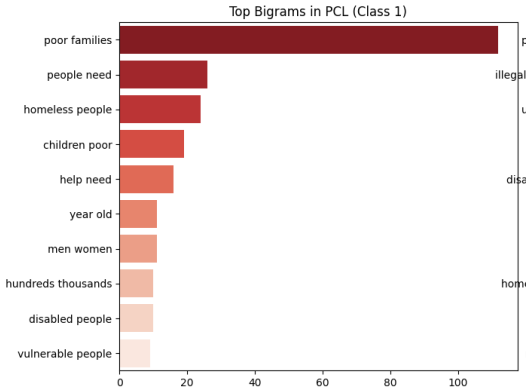
If we assume that PCL consists of 2-4 and Non-PCL is 0-1, there are 90% in Non-PCL and 10% as PCL.

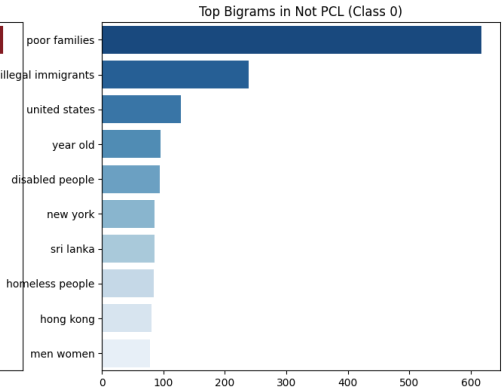
2.1.2 Impact Statement

The length distribution of my model shows that most of our paragraphs aren't extremely long, hence we may benefit for truncating our data. That being said, we would want to experiment with different token lengths (perhaps 256) as we will want to keep the context in longer paragraphs such as the paragraph of 909 tokens. Due to the classes imbalance, standard accuracy will not be a good metric to measure our model. Hence, we may need to resort to F1-score or maybe over-sampling in training

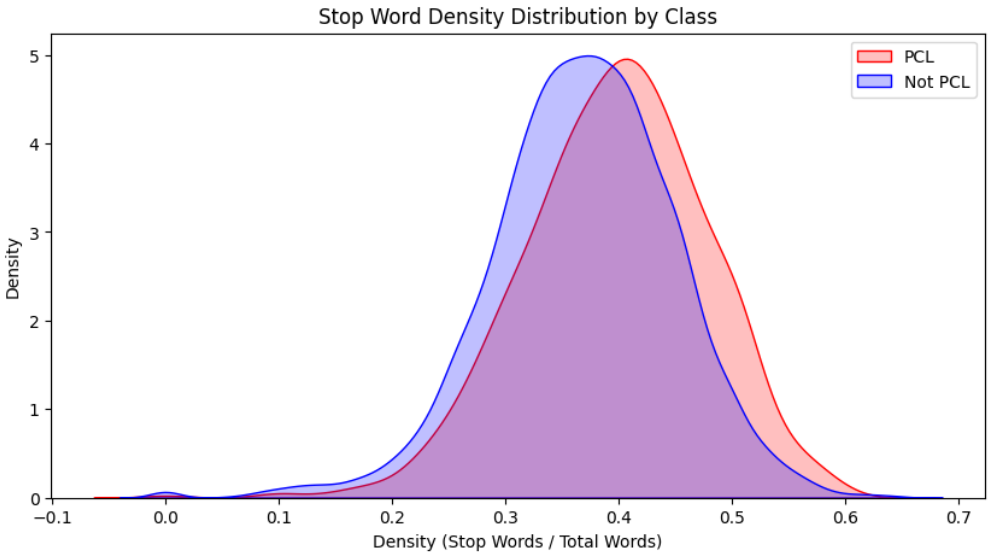
2.2 Lexical Analysis

2.2.1 Ngrams





2.2.2 Stop words



2.2.3 Word Cloud

Table 2: Top 10 High-Frequency Words in PCL vs. Non-PCL Samples

Word	PCL Count	Non-PCL Count	PCL Freq %
people	212	1286	26.70
need	197	874	24.81
poor	180	762	22.67
children	166	639	20.91
families	159	901	20.03
said	156	1593	19.65
homeless	148	674	18.64
women	107	1193	13.48
help	106	312	13.35
disabled	76	827	9.57

2.2.4 Analysis

1. **Bigrams**: PCL samples are dominated by phrases such as "poor families" and "people need", slightly more descriptive compared to Non-PCL where we see names of places such as "Sri Lanka" get seen frequently.
2. **Stop word density**: Both classes PCL and Non-PCL have similar density distributions (around 0.4)
3. **Word Cloud**: We notices that words that are frequent in PCL are also frequent in Non-PCL, such as people. however we get some disparity in words such as "said" and "women".

2.2.5 Impact

1. **Bigrams**: The condescending language is within the relationship between the subject and the accompanying evaluative language. This does suggest that a BERT-like model maybe be a good starting point due to its ability to use bidirectional context.

2. **Stop word density**: We see that nearly half the text consists of filler tokens. We may have to consider not fully removing stop words as they themselves might have subtle linguistic markers of PCL
3. **Word Cloud** The overlap of community keywords in both classes does show that keyword-matching would be ineffective.

3 Stage 3: Baseline Model and proposing a novel approach

3.1 Proposed Solution

My strategy to surpass the RoBERTa-base baseline is a multi-step pipeline, that uses BERT-Large-Uncased architecture, data augmentation and cost-sensitive learning.

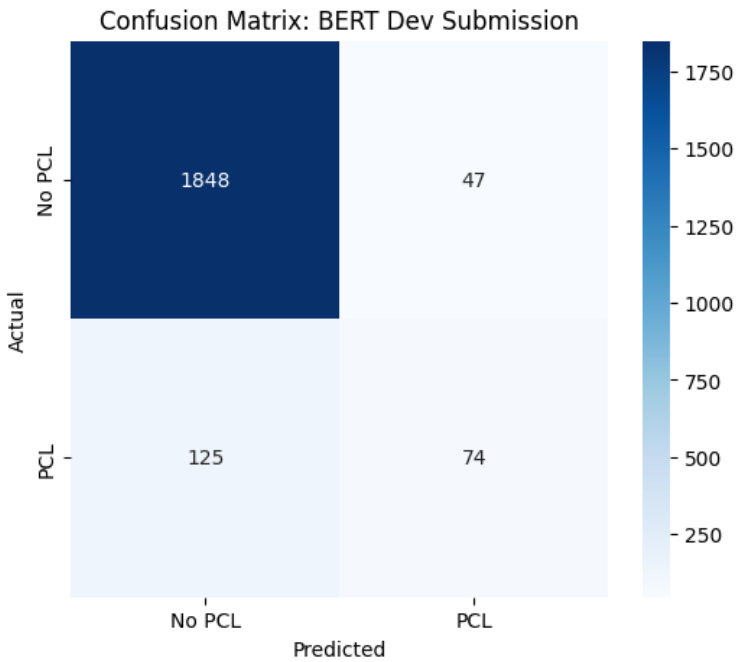
This strategy includes:

1. Strategic Data Modification (Augmentation). To address the class imbalance, I integrated a few data augmentation techniques.
   - Back-Translation: Converting PCL paragraphs from English to German then back to English, to generate more data
   - Random deletion: I delete a random word on some PCL data points. In hopes of not losing the semantic meaning, we may need to adjust how frequent and how much is deleted
2. Weighted loss function: I aim a weighted loss function that heavily penalises misclassifications of minority classes, i.e. the PCL class.
3. The model I will be initially experimenting with is BERT-Large-Uncased with a low learning rate due to its larger architecture with 340m parameters compared to 125m in RoBERTa-base.

3.2 Expected Outcome

The expected outcome is a model that demonstrates a significantly higher Recall for the PCL class than the baseline, successfully identifying subtle condescension that the standard RoBERTa model ignored. By addressing the "small N" overfitting problem through synthetic data and mathematically balancing the loss function, the model is expected to comfortably surpass the baseline 0.48 F1 score on the official dev-set.

4 Stage 5.2: Local Evaluation and Error Anaalysis



The model achieved an overall accuracy score of 0.92 (with augmentation), as discussed before is a misleading statistic. Our plan to identify PCL shows significant room for improvement with an F1-score of of 0.46, lower than the baseline

4.1 Model Failures

4.2 Tables

Table 3: Final Evaluation Results: BERT with Weighted Trainer (With Data Augmentation)

Class	**Precision**	**Recall**	**F1-Score**	**Support**
No PCL	0.94	0.98	0.96	1895
PCL	0.61	0.37	0.46	199
Accuracy			**0.92**	2094
Macro Avg	0.77	0.67	0.71	2094
Weighted Avg	0.91	0.92	0.91	2094

Table 4: Final Evaluation Results: BERT with Weighted Trainer (Without Data Augmentation)

Class	**Precision**	**Recall**	**F1-Score**	**Support**
No PCL	0.95	0.97	0.96	1895
PCL	0.64	0.51	0.57	199
Accuracy			**0.93**	2094
Macro Avg	0.80	0.74	0.76	2094
Weighted Avg	0.92	0.93	0.92	2094

- False negatives: These are the most critical errors. we observe that the model failed to identify 63% of PCL instances (recall=0.37) while No PCL achieved an F1 score of 0.96. This demonstrates the BERT's actions to be overly conservative, perhaps due to the subtlety of patronising language and lack of more obvious keywords.
- False positives: The model is relatively precise achieving approximately 61% for PCL. We occasionally observe that the model flags neutral text. For example: *"So, let the NPP government appointees look down upon the hardworking and selfless men and women at their own peril"*. Using "look down upon" may have cause the model to flag this

4.2.1 BERT

Bert-Large is powerful especially with its large amount of parameters. It uses absolute position embeddings. However, it did show its limitations due to its simplicity. The false positive example spoken about above does demonstrate that BERT is explicitly looking for literal phrase semantic matching rather than the hidden relationship between the subject and adjective.

4.2.2 Effect of Data Augmentation

I suspected that the data augmentation was affecting the model's ability to understand the specific semantics of PCL. To test this, I compared the initial BERT model using data augmentation (Back-translation and Random Deletion) against a BERT model using only a Weighted Trainer.

Table 5: Comparison: Data Augmentation(DT) vs. Weighted Training(WT) (BERT)

Experimental Setup	**Precision**	**Recall**	**F1-Score**
BERT + WT + DA	0.61	0.37	0.46
BERT + WT	**0.64**	**0.51**	**0.57**
Improvement	**+4.9%**	**+37.8%**	**+23.9%**

We notice that we have surpassed the baseline F1-score of 0.49 by using only Weighted Training (weighted cross entropy loss to achieve 0.57 and surpassing the data augmentation by 23.9% relatively. We also see a huge boost in Recall, noticing significant improvement. in recall. The sole purpose of the data augmentation was to artificially inflate the data set and learn the small dataset. We have now understood that perhaps back-translation can be too drastic and lose the subtle relationship we aim to learn

Note that when I initally wrote up Stage 5.2, I did not experiment with removing Data Augmentation hence why, I examine that experiment in the other sections.